An Elementary Uniform Spectral Gap for the Syracuse Transfer Operator

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https://github.com/m4cd4r4/collatz-cycle-certificate

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Abstract

Let T_k be the transfer operator of the Syracuse map $\mathrm{Syr}(n) = (3n+1)/2^{v_2(3n+1)}$ on the $N = 2^{k-1}$ odd residues modulo 2^k, averaging over the 2^k lifts of a residue. We prove a uniform spectral gap: for every $k \geq 3$,

$$|\lambda_2(T_k)| \;\leq\; \mathrm{cert}(k) \;\leq\; 2^{-3/2} + 2^{-1} \;=\; 0.8535\ldots \;<\; 1,$$

where $\mathrm{cert}(k)$ is an explicit weighted row-sum certificate over a 2-adic frequency grading. The proof is entirely elementary: finite cyclic 2-group arithmetic (three uniform-fiber lemmas sharing one engine, "3 is a unit modulo 2^k"), a finite geometric series, an integer collision count, and a single application of the Cauchy–Schwarz inequality. A sharper per-level bound on the defect covector improves the constant to 0.656; the measured values are $\mathrm{cert}(k) \approx 0.6345$ and $|\lambda_2| \approx 0.27$. The elementary core of the proof is machine-checked in Lean 4 (Mathlib, sorry-free).

Scope. This is a theorem about the Markov model, not about Collatz orbits. In particular it does *not* rule out Collatz cycles: the analogous $3x-1$ operator passes the identical certificate even more strongly, yet $3x-1$ has non-trivial cycles. An earlier version of this project claimed a cycle-elimination corollary; that claim was withdrawn (Section 9).

Keywords: Syracuse map, transfer operator, spectral gap, 2-adic analysis.

1 Introduction

For odd n, the Syracuse map is $\mathrm{Syr}(n) = (3n+1)/2^{v_2(3n+1)}$, where v_2 is the 2-adic valuation. Reducing modulo 2^k gives a natural Markov chain on the $N = 2^{k-1}$ odd residues: from a residue r, move to $\mathrm{Syr}(r + m2^k) \bmod 2^k$ with m uniform on $[0, 2^k)$. Its transition matrix T_k (column-stochastic) is the finite-scale transfer operator of the map; the mixing of this family is the standard probabilistic model for the equidistribution of Syracuse trajectories modulo powers of two, going back to the heuristics surveyed in [1] and quantified in Tao's almost-all result [2].

Our main result is a uniform, explicit, and elementarily proved spectral gap for this family.

*With Claude (Anthropic): Opus 4.8 research sessions and a Fable 5 adversarial review, refereeing, and consolidation pass. All proofs are elementary and every numerical claim is reproducible from the repository.

Theorem 1.1 (Uniform spectral gap). *For every $k \geq 3$,*

$$|\lambda_2(T_k)| \;\leq\; \mathrm{cert}(k) \;:=\; \max_{0\leq a\leq k-2} \sum_{b=0}^{k-2} Q_k[a,b]\, 2^{a-b} \;\leq\; 2^{-3/2} + 2^{-1} \;<\; 1,$$

where λ_2 is the second-largest eigenvalue in modulus and $Q_k[a,b] = \|P_a U_k P_b\|_2$ is the matrix of block operator norms of the Koopman adjoint $U_k = T_k^\top$ over the 2-adic frequency grading of Section 2.

The interest of the theorem is threefold. First, uniformity: the bound is independent of k, while the state space grows as 2^{k-1}. Second, the constant is explicit and the true values sit far below it ($|\lambda_2| \approx 0.27$ for all tested k), so the certificate has a large margin. Third, and mainly, the proof is elementary from end to end. What looks like it should require Gauss-sum estimates reduces to three instances of one fact (multiplication by a unit has uniform fibers on a cyclic 2-group), one finite geometric series, one integer collision count with a two-line injectivity argument, and one Cauchy–Schwarz application. No analytic number theory, no perturbation theory, and no computer-assisted spectral certification enters the proof; numerics appear only as consistency checks.

We are explicit about history because this project previously overclaimed twice, and both corrections shaped the final form. (i) A March 2026 draft asserted the gap via an eigenvalue perturbation telescope with a "bounded condition number"; an adversarial audit (Session 39) refuted this: the eigenvector condition number grows like 10^k, and the published inductive inequality fails numerically at every odd $k \geq 7$. The present proof replaces eigenvalue perturbation with norm inequalities that are condition-number-free. (ii) The certificate was then presented as eliminating non-trivial Collatz cycles. That inference is false and was withdrawn; Section 9 records the decisive control experiment. What survives is the operator theorem itself.

Proof architecture

1. **Spectral reduction** (Section 3): $|\lambda_2(T_k)| \leq \rho(Q_k) \leq \mathrm{cert}(k)$, by a one-sided Perron split, a level-energy majorisation applied to a compression, and Gelfand's formula. A subtlety repaired late in the project: T_k is *not* doubly stochastic and its stationary distribution is *not* exactly uniform, so the argument must (and does) avoid both.

2. **Block structure** (Section 4): $U_k = U_{\mathrm{clean}} + D$ where U_{clean} is strictly upper-triangular in the level grading with exactly computable block norms $\|P_a U_{\mathrm{clean}} P_b\| = 2^{-(b-a)/2}$ (Lemma A), and D is a rank-1 defect supported at the single residue $r^* = -3^{-1} \bmod 2^k$.

3. **Defect mass** (Section 5): the defect covector c satisfies $\|c\|_2 \leq \sqrt{3}\, 2^{-k/2}$, by an integer collision count (Lemma B).

4. **Assembly** (Section 6): each weighted row sum is bounded by the envelope $f(e) = \sum_{d=1}^{e-2} 2^{-3d/2} + 2^{-(e-1)/2}$ with $e = k - a$, whose maximum is $f(3) = 2^{-3/2} + 2^{-1}$. The only analytic input is Cauchy–Schwarz; crucially, the upper cascade *truncates* at the top rows, which is what makes the L^2 bound sufficient.

Section 7 records the sharper per-level defect bound (Lemma C, via an elementary frequency-doubling identity), which improves the constant to 0.656 but is not needed for Theorem 1.1. Section 8 describes the Lean 4 formalisation of the elementary core. Section 9 delimits scope.

2 The operator and the frequency grading

Definition 2.1. *Fix $k \geq 3$ and let $N = 2^{k-1}$. The state space is the set of odd residues modulo 2^k. The chain moves from r to $\mathrm{Syr}(r+m2^k) \bmod 2^k$ with m uniform on $\{0, \dots, 2^k - 1\}$; $T_k[s', s]$ is this transition probability, and $U = U_k := T_k^\top$ is the Koopman (expectation) operator, $(Ug)(r) = \mathbb{E}_m\, g\big(\mathrm{Syr}(r + m2^k)\big)$.*

T_k is column-stochastic by construction, so $\mathbf{1}^\top T_k = \mathbf{1}^\top$ and $\rho(T_k) = 1$.

Remark 2.2 (T_k is not doubly stochastic)**.** *The row sums of T_k are* not *identically 1: numerically $\|T_k\mathbf{1} - \mathbf{1}\|_2 \approx 0.764 \cdot 2^{-k/2}$, another face of the r^* defect of Section 4. Consequently the stationary distribution is only approximately uniform, U does not preserve the mean-zero subspace, and identities such as $T_r^p = T_k^p - N^{-1}J$ (used in an earlier write-up) are false. Nothing below uses uniformity of the stationary distribution; the reduction in Section 3 is deliberately one-sided.*

Definition 2.3 (Characters and levels)**.** *Let $\omega = e^{2\pi i/2^k}$ and, for $\xi \in [0, 2^{k-1})$, let $\chi_\xi(r) = \omega^{\xi r}/\sqrt{N}$ on odd r. On odd arguments $\chi_{\xi+2^{k-1}} = -\chi_\xi$, so this range selects one representative per proportionality class, and the χ_ξ form an orthonormal basis of $\mathbb{C}^N$ (orthogonality is the $m = k$ case of Lemma 4.3 below). χ_0 is the constant (Perron) vector. The* level *of $\xi \neq 0$ is $v_2(\xi) \in \{0, \dots, k-2\}$; P_a denotes the orthogonal projection onto $W_a = \mathrm{span}\{\chi_\xi : v_2(\xi) = a\}$, of dimension $d_a = 2^{k-2-a}$. The mean-zero space is $V = W_0 \oplus \cdots \oplus W_{k-2}$, an orthogonal decomposition.*

3 The spectral reduction

Proposition 3.1. *Let $Q = Q_k$ with $Q[a, b] = \|P_a U P_b\|_2$. Then $|\lambda_2(T_k)| \leq \rho(Q) \leq \mathrm{cert}(k)$.*

Proof. (i) Perron split. Since $\mathbf{1}^\top T = \mathbf{1}^\top$, the mean-zero space $V = \ker \mathbf{1}^\top = \mathbf{1}^\perp$ is T-invariant. In a basis adapted to $\mathbb{C}^N = \mathrm{span}(\mathbf{1}) \oplus V$, T is block-triangular $\left(\begin{smallmatrix} 1 & 0 \\ d & A \end{smallmatrix}\right)$ with $A = T|_V$ (the vector d, the V-component of $T\mathbf{1}$, is nonzero by Remark 2.2 but harmless). The characteristic polynomial factors, so $\mathrm{spec}(T) = \{1\} \uplus \mathrm{spec}(A)$ and $|\lambda_2(T)| \leq \rho(A)$.

(ii) Adjoint and Gelfand. $A^* = (P_V T P_V)^*|_V = P_V U P_V =: U_V$, the compression of U to V (V need not be U-invariant; only the compression is used), and $\rho(A) = \rho(U_V)$. Gelfand's formula $\rho(U_V) \leq \|U_V^p\|_2^{1/p}$ holds for every matrix and every p; no normality or diagonalisability enters. This is the step that makes the argument immune to the 10^k eigenvector condition numbers that invalidated the earlier perturbative route.

(iii) Level majorisation. For $g \in V$ let $\alpha(g)_a := \|P_a g\|_2$. Since $P_a P_V = P_a$, the triangle inequality gives, for each level a',

$$\|P_{a'} U_V g\| = \Big\|\sum_b P_{a'} U P_b g\Big\| \leq \sum_b Q[a', b]\, \alpha(g)_b = (Q\, \alpha(g))_{a'},$$

i.e. $\alpha(U_V g) \le Q\,\alpha(g)$ entrywise. As $Q \ge 0$ this iterates: $\alpha(U_V^p g) \le Q^p \alpha(g)$. The levels are orthogonal and exhaust V, so $\|U_V^p g\|_2 = \|\alpha(U_V^p g)\|_2 \le \|Q^p\|_2 \|g\|_2$, whence $\rho(U_V) \le \lim_p \|Q^p\|^{1/p} = \rho(Q)$.

(iv) Weighted row sum. For the diagonal similarity $S = \mathrm{diag}(2^0, \dots, 2^{k-2})$ and the nonnegative matrix Q, $\rho(Q) = \rho(SQS^{-1}) \le \|SQS^{-1}\|_\infty = \max_a \sum_b Q[a,b]\,2^{a-b} = \mathrm{cert}(k)$. □

4 Block structure: three instances of one engine

Throughout, x is odd, $v(r) := v_2(3r+1) \ge 1$, and 3^{-1} denotes the inverse of 3 in the relevant $\mathbb{Z}/2^m$.

4.1 Coset uniformity and the splitting $U = U_{\mathrm{clean}} + D$

Lemma 4.1 (CU)**.** *Let x be odd with $v := v_2(3x+1) < K$, and $q_0 := (3x+1)/2^v$ (odd). Then for every m,*

$$v_2\big(3(x+m2^K)+1\big) = v, \qquad \mathrm{Syr}(x+m2^K) = q_0 + 3m\,2^{K-v},$$

and as m ranges over $\mathbb{Z}/2^K$ the residue $\mathrm{Syr}(x+m2^K) \bmod 2^K$ is uniform on the coset $q_0 + 2^{K-v}(\mathbb{Z}/2^K)$ of size 2^v, each value attained exactly 2^{K-v} times.

Proof. $3(x+m2^K)+1 = 2^v q_0 + 3m2^K = 2^v(q_0 + 3m\,2^{K-v})$, and the bracket is odd because q_0 is odd and $K - v \ge 1$; this freezes the valuation and gives the affine formula. For the count: $m \mapsto 3 \cdot 2^{K-v} m$ is an endomorphism of $\mathbb{Z}/2^K$ whose image is the subgroup $2^{K-v}(\mathbb{Z}/2^K)$ of order 2^v (3 is a unit) and whose kernel has order 2^{K-v}; every image point has exactly 2^{K-v} preimages, and adding q_0 translates the image to the coset. □

Apply Lemma 4.1 with $K = k$. For every row r with $v(r) < k$, averaging χ_η over the uniform coset collapses to a geometric sum which is 2^v or 0 according to whether $2^v \mid \eta$, yielding the masked-phase formula

$$(U\chi_\eta)(r) = \mathbf{1}[\,v(r) \le v_2(\eta)\,]\,\omega^{\eta\, q(r)}, \qquad q(r) := (3r+1)/2^{v(r)}, \qquad (v(r) < k). \tag{1}$$

Exactly one odd residue fails the hypothesis: $2^k \mid 3r+1$ has the unique odd solution $r^* = -3^{-1} \bmod 2^k = (4^{\lceil k/2 \rceil} - 1)/3$, where $3r^* + 1 = 2^{2\lceil k/2 \rceil}$ exactly. Define U_{clean} by (1) with the r^*-row set to zero, and $D := U - U_{\mathrm{clean}}$. Then:

- **(R1)** $P_a U_{\mathrm{clean}} P_b = 0$ for $a \ge b$ (strict upper-triangularity; a by-product of the survivor analysis below);
- **(R2)** $D = e_{r^*} c^*$ is rank 1, where c is the probability vector $c[t] = 2^{-k}\#\{m : \mathrm{Syr}(r^* + m2^k) \equiv t \bmod 2^k\}$; hence $\|P_a D P_b\| = u_a v_b$ with $u_a := \|P_a e_{r^*}\|$, $v_b := \|P_b c\|$;
- **(R3)** $u_a = 2^{-(a+1)/2}$ exactly: any state unit vector has flat character energy $1/N$ per frequency, so $u_a^2 = d_a/N = 2^{-1-a}$.

By the triangle inequality, for all a, b:

$$Q[a,b] \;\le\; \|P_a U_{\mathrm{clean}} P_b\| + u_a v_b. \tag{2}$$

4.2 The two remaining elementary inputs for Lemma A

Lemma 4.2 (SB: shell bijection)**.** *Fix* $1 \le j \le k-2$*. The map* $\psi(r) := (3r+1)/2^j \bmod 2^{k-j}$ *is a bijection from the shell* $S_j := \{r \text{ odd}, 0 < r < 2^k,\ v(r) = j\}$ *onto the odd residues modulo* 2^{k-j}*.*

Proof. Injectivity: $\psi(r) = \psi(r')$ gives $3r+1 \equiv 3r'+1 \pmod{2^k}$, so $r = r'$ (3 a unit, both in $[0, 2^k)$). Cardinality: $v(r) = j$ pins r to a single (odd) residue class modulo 2^{j+1}, so $|S_j| = 2^{k-j-1}$, which equals the number of odd residues modulo 2^{k-j}; injective plus equal finite cardinality gives bijective, with multiplicity exactly one. □

Lemma 4.3 (S4: odd geometric sum)**.** *For* $m \ge 1$ *and* $S^{\mathrm{odd}}(\alpha, m) := \sum_{q \text{ odd}, 0 \le q < 2^m} \omega^{\alpha q}$*: if* $2^{k-1} \mid \alpha$ *then* $|S^{\mathrm{odd}}| = 2^{m-1}$ *(full); otherwise* $S^{\mathrm{odd}} = 0$ *if and only if* $k - m \le v_2(\alpha) \le k-2$ *(the dead band); below the band it is a nonzero partial sum.*

Proof. Substitute $q = 2\ell + 1$ and sum the geometric series with ratio $\omega^{2\alpha}$. □

4.3 Lemma A: the clean upper cascade is a scaled isometry

Theorem 4.4 (Lemma A)**.** *For all* k *and all* $0 \le a < b \le k-2$*, with* $d := b - a$*, the block* $B = P_a U_{\mathrm{clean}} P_b$ *satisfies* $B^*B = 2^{-d} I$*; in particular* $\|P_a U_{\mathrm{clean}} P_b\|_2 = 2^{-d/2}$ *exactly. Moreover* $P_a U_{\mathrm{clean}} P_b = 0$ *for* $a \ge b$ *(R1).*

Proof (survivor analysis). The matrix entry of U_{clean} between source χ_η $(v_2(\eta) = b)$ and target χ_ξ $(v_2(\xi) = a)$ is, up to the $1/N$ normalisation, $\hat{U}(\eta, \xi) = \sum_{r, 1 \le v(r) \le b} \omega^{\eta q(r) - \xi r}$. Group by the shell $j = v(r)$. On shell j, $r = 3^{-1}(2^j q - 1)$ with $q = q(r)$, so the phase is $q\,\alpha_j + \xi 3^{-1}$ with $\alpha_j := \eta - \xi\, 2^j 3^{-1}$. The valuations are

$$v_2(\alpha_j) = a + j \ (j < d), \qquad v_2(\alpha_j) = b \ (j > d), \qquad v_2(\alpha_j) \ge b \ (j = d),$$

and in every case $v_2(\alpha_j) \ge j$, so $\omega^{q\alpha_j}$ depends only on $q \bmod 2^{k-j}$. By Lemma 4.2 the shell sum is exactly $\omega^{\xi 3^{-1}} S^{\mathrm{odd}}(\alpha_j, k-j)$, and by Lemma 4.3 every shell lands in the dead band $[j, k-2]$ *except* possibly $j = d$, which survives if and only if $\alpha_d \in \{0, 2^{k-1}\}$, with full modulus 2^{k-d-1}. Counting: each η has exactly 2^d owners ξ (two residue classes modulo 2^{k-d}, each contributing 2^{d-1} values of valuation a in $[0, 2^{k-1})$), giving the diagonal $(B^*B)[\eta, \eta] = N^{-2} \cdot 2^d \cdot (2^{k-d-1})^2 = 2^{-d}$; and for fixed ξ exactly one of the two candidate sources lies in the index range $[0, 2^{k-1})$, so distinct columns have disjoint row support and the off-diagonal entries vanish. For $a \ge b$ the same computation gives $v_2(\alpha_j) = b \in [j, k-2]$ for every shell $j \le b$, so every shell is annihilated: this is (R1). □

5 Lemma B: the defect mass, by an integer collision count

Theorem 5.1 (Lemma B)**.** $\displaystyle\sum_{b=0}^{k-2} v_b^2 \ \le\ \|c\|_2^2 \ =\ \frac{\mathrm{coll}(k)}{4^k} \ \le\ \frac{3 \cdot 2^k}{4^k},$ *i.e.* $\|v\|_2 \le \sqrt{3}\, 2^{-k/2}$*, where* $\mathrm{coll}(k) = \sum_t \mathrm{cf}[t]^2$ *and* $\mathrm{cf}[t] = \#\{m \in [0, 2^k) : \mathrm{Syr}(r^* + m2^k) \equiv t \bmod 2^k\}$*.*

Proof sketch (full details in the repository). Since $3r^* + 1 = 2^{2\lceil k/2\rceil}$, the fiber is $\mathrm{Syr}(r^* + m2^k) = \mathrm{oddpart}(a+3m) \bmod 2^k$ with $a = 2^{2\lceil k/2\rceil - k} \in \{1, 2\}$, an arithmetic progression of length 2^k in the integers. Decompose it by 2-adic shells $A_j = \{x = a + 3m : v_2(x) = j\}$, of sizes $|A_j| = 2^{k-1-j}$ $(j \le k-1)$ plus one top atom.

FACT 1 (per-shell injectivity). On each shell, $x \mapsto (x/2^j) \bmod 2^k$ is injective: if $x = 2^j u$, $x' = 2^j u'$ with $u \equiv u' \pmod{2^k}$, then $3(m - m') = 2^j(u - u') \equiv 0 \pmod{2^{j+k}}$; as $\gcd(2, 3) = 1$ this forces $2^{j+k} \mid m - m'$, and $|m - m'| < 2^k$ gives $m = m'$. Hence the fiber count is 0/1 per shell, and $\mathrm{coll} = \sum_{j,j'} |R_j \cap R_{j'}|$ over the shell images R_j.

The diagonal contributes exactly 2^k. The cross terms obey $\sum_{j<j'} |R_j \cap R_{j'}| \le \sum_{j' \ge 2} (j' - 1)2^{k-1-j'} + (k-1) = 2^{k-1} - 1$ (the closed form includes the top atom's at most $k-1$ partners, covered by the truncation slack of the geometric series). Hence $\mathrm{coll} \le 2^k + 2(2^{k-1} - 1) < 3 \cdot 2^k$. The sharp constant is $\mathrm{coll}/2^k \to 31/12$, comfortably inside 3. Finally $\sum_b v_b^2 = \|c\|^2 - 1/N \le \|c\|^2$, the dropped term being the Perron coefficient of the probability vector c. □

6 The assembly

Proof of Theorem 1.1. Fix a row $a \le k-2$ and set $e := k - a \ge 2$. Write the weight as $2^{a-b} = 2^a (1/2)^b$. By (2), Theorem 4.4 and (R1),

$$\sum_b Q[a, b]\, 2^{a-b} \;\le\; \underbrace{\sum_{d=1}^{e-2} 2^{-d/2}\, 2^{-d}}_{\text{clean cascade, truncated}} \;+\; \underbrace{u_a 2^a \sum_{b=0}^{k-2} v_b\, 2^{-b}}_{\text{defect, all } b}.$$

Two remarks. The clean sum *truncates* at $d \le e-2$ because $b \le k-2$; at the top row $a = k-2$ it is empty. The defect is summed over *all* b, which absorbs its contribution to the upper blocks with no separate estimate.

For the defect factor, Cauchy–Schwarz against the geometric weight and Theorem 5.1 give, with no information about the profile of (v_b),

$$\sum_{b=0}^{k-2} v_b\, 2^{-b} \;\le\; \|v\|_2 \Big(\sum_{b \ge 0} 4^{-b}\Big)^{1/2} \;\le\; \sqrt{3}\, 2^{-k/2} \cdot \tfrac{2}{\sqrt{3}} \;=\; 2^{1-k/2},$$

and with $u_a 2^a = 2^{(a-1)/2}$ (R3) the defect term is at most $2^{(a-1)/2}\, 2^{1-k/2} = 2^{-(e-1)/2}$. Hence

$$\sum_b Q[a, b]\, 2^{a-b} \;\le\; f(e) \;:=\; \sum_{d=1}^{e-2} 2^{-3d/2} + 2^{-(e-1)/2}.$$

The difference $f(e+1) - f(e) = 2^{-(e-1)/2}\big(2^{-(e-1)} - (1 - 2^{-1/2})\big)$ is positive only at $e = 2$, so f increases from $f(2) = 2^{-1/2}$ to its maximum $f(3) = 2^{-3/2} + 2^{-1} = 0.8535\ldots$ and decreases thereafter toward $G_{\mathrm{up}} = (2^{3/2} - 1)^{-1} = 0.5469\ldots$. Therefore $\mathrm{cert}(k) \le f(3) < 1$ for every $k \ge 3$, and Proposition 3.1 completes the proof. □

Remark 6.1 (Why truncation matters)**.** *Without the truncation one would bound the row sum by $G_{\mathrm{up}} + 2^{-1/2} = 1.25 > 1$ and wrongly conclude that the L^2 defect bound is insufficient (an earlier version of the assembly did exactly this and introduced the per-level Lemma C as a necessity). Row-wise, the two terms of f trade off: rows with a long cascade have a small defect weight and vice versa, with the worst case at $e = 3$.*

Remark 6.2 (Sharpness and measured values)**.** *The envelope is asymptotically sharp at the bottom row ($e = k$), where both the true row sum and $f(k)$ converge to G_{up}; the row that sets the constant, $e = 3$, has a wide margin (true value ≈ 0.63 against $f(3) = 0.854$). Ground-truth values (dense operator, both parities; reproducible with* `fable_assembly_check.py`*):*

k	4	6	8	10	12	13
$\mathrm{cert}(k)$	0.6117	0.6338	0.6347	0.6345	0.6345	0.6344
$\rho(Q_k)$	0.5020	0.5535	0.5661	0.5673	0.5656	0.5644
$\lVert c\rVert_2\, 2^{k/2}$	1.6202	1.6105	1.6081	1.6075	1.6073	1.6072

All row bounds $R_a \le f(k-a)$ hold with margin at every tested k, and the rank-1 factorisation $Q[a, b] = u_a v_b$ on $a \ge b$ is machine-exact.

7 The per-level sharpening (Lemma C)

Theorem 1.1 needs only the L^2 mass of the defect. The per-level structure is nevertheless elementary and sharp, and improves the constant.

Theorem 7.1 (Lemma C)**.** *$v_b \le \frac{3}{4} 2^{-b} 2^{-k/2}$ for all k and $0 \le b \le k-2$.*

The proof (in the repository) rests on a frequency-doubling identity for the fiber exponential sum $S_k(\xi) = \sum_t \mathrm{cf}[t]\, e^{-2\pi i \xi t/2^k}$:

Lemma 7.2 (H: homometry)**.** *For $k \ge 2$ and $2\xi \notin \{0, 2^{k-1}\} \bmod 2^k$: $S_k(2\xi) = S_{k-1}(\xi)$ exactly.*

Proof idea. Split the fiber sum by the parity of m. One parity makes $3m + a$ odd and contributes a geometric series that vanishes away from the two excluded frequencies; the other parity halves the scale and reproduces the $(k-1)$-fiber exactly (the parity flip swaps the offsets $a = 1 \leftrightarrow 2$, matching the parity of $k-1$). □

Iterating Lemma 7.2 reduces the level-b energy to a single scale $p = k - b$, where a Parseval identity converts it into the half-shift autocorrelation $\mathrm{coll}(p) - A_p$, which the shell decomposition of Theorem 5.1 bounds by $3 \cdot 2^{p-1} - 2$ (the $j = 0$ shell cancels identically from the autocorrelation). Feeding Theorem 7.1 through the truncated envelope of Section 6 replaces $2^{-(e-1)/2}$ by $2^{-(e+1)/2}$ and yields

$$\mathrm{cert}(k) \;\le\; \max_{e\ge 2}\Big[\sum_{d=1}^{e-2} 2^{-3d/2} + 2^{-(e+1)/2}\Big] \;=\; 0.6559\ldots \qquad \text{(maximum at } e = 4\text{)},$$

remarkably close to the measured 0.6345.

8 Lean formalisation

The elementary core of the proof is machine-checked in Lean 4 against Mathlib (v4.27.0), in the repository module `lean/GapCertificate.lean` (15 theorems, no `sorry`, no new axioms):

- the shared divisibility engine ($2^N \mid 3(m - m')$ in a 2^N-window forces $m = m'$), and as corollaries FACT 1 of Theorem 5.1 and the injectivity half of Lemma 4.2;
- the affine step of Lemma 4.1 (`cu_valuation_frozen`, `cu_syracuse_affine`);
- the envelope analysis of Section 6: $f(e) \le f(3) < 1$;
- the Cauchy–Schwarz defect bound and the abstract row-sum assembly (`assembly_row_bound`, `certificate_lt_one`): given the block bounds of Sections 4–5 as hypotheses, every weighted row sum is $\le f(k - a) < 1$.

Two formalisation dividends: FACT 1, parametrised by the lift index, needs no modulo-3 argument at all; and the identity $(2s^2)^{a+1} = 1$ for $s = 2^{-1/2}$ keeps every exponent in $\mathbb{N}$, so the formal assembly involves no real exponentiation. Not yet formalised: the operator chain of Proposition 3.1, the fiber count of Lemma 4.1, the counting halves of Lemmas 4.2 and B, and Lemma C.

9 What this theorem does and does not say

Theorem 1.1 concerns the averaged Markov model: it says the mod-2^k chain mixes uniformly fast in k. It says nothing about individual deterministic orbits.

The withdrawn cycle claim. An earlier version of this project asserted that the uniform gap eliminates non-trivial Collatz cycles. The inference is false, by a control experiment: the $3x - 1$ map (equivalently, Collatz on negative integers) has non-trivial cycles ($\{5, 7\}$, $\{17, 25, 37, 55, 41, 61, 91, 47, 71, 53, 79, 59, 89, 33, 49, 73\}$, $\ldots$), yet its transfer operator passes the identical certificate, with an even stronger measured gap. The averaging over 2^k lifts that defines T_k erases exactly the deterministic orbit structure a cycle lives in, in the same way that the doubling map is mixing while having dense periodic orbits. Ruling out Collatz cycles requires instruments sensitive to specific orbits (heights, linear forms in logarithms), not the spectrum of the averaged operator. Likewise the theorem has no bearing on divergent trajectories; the determinism-versus-probability gap quantified in the project's audit (a large-deviation shortfall of roughly a factor 12 for every counting-based route) remains the actual wall, consistent with the state of the art after [2].

What the theorem does provide: a natural family of arithmetically structured, non-normal, non-doubly-stochastic operators of exponentially growing dimension with a uniform, explicit, elementarily provable spectral gap, whose sharp structure ($B^*B = 2^{-d}I$ blocks, a rank-1 defect at the base-4 repunit r^*, the $31/12$ collision constant) is completely understood.

Reproducibility

All claims are reproducible from `https://github.com/m4cd4r4/collatz-cycle-certificate`: the proof documents (`THEOREM.md` and the lemma files), the numerical ground-truth checks (`fable_assembly_check.py`, `verify_assembly.py`, `audit_halfshift_s4.py`, `lemmaB_fact1_rig` the $3x-1$ control (`probe_cycle_link.py`, `CYCLE_CLAIM_REFUTED.md`), and the Lean project (`lean/`).

References

[1] J. C. Lagarias (ed.), *The Ultimate Challenge: The $3x+1$ Problem*, American Mathematical Society, 2010.

[2] T. Tao, *Almost all orbits of the Collatz map attain almost bounded values*, Forum of Mathematics, Pi 10 (2022), e12. arXiv:1909.03562.

[3] L. Mori, *$C^*(T_1, T_2)$ irreducible on $\ell^2(\mathbb{N})$ iff Collatz*, arXiv:2411.08084 (2024).

[4] The mathlib Community, *The Lean mathematical library*, CPP 2020.